



Faculty of Science

SQL4 vs. SQLite

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Advanced Programming, DIKU, 2026-10-20



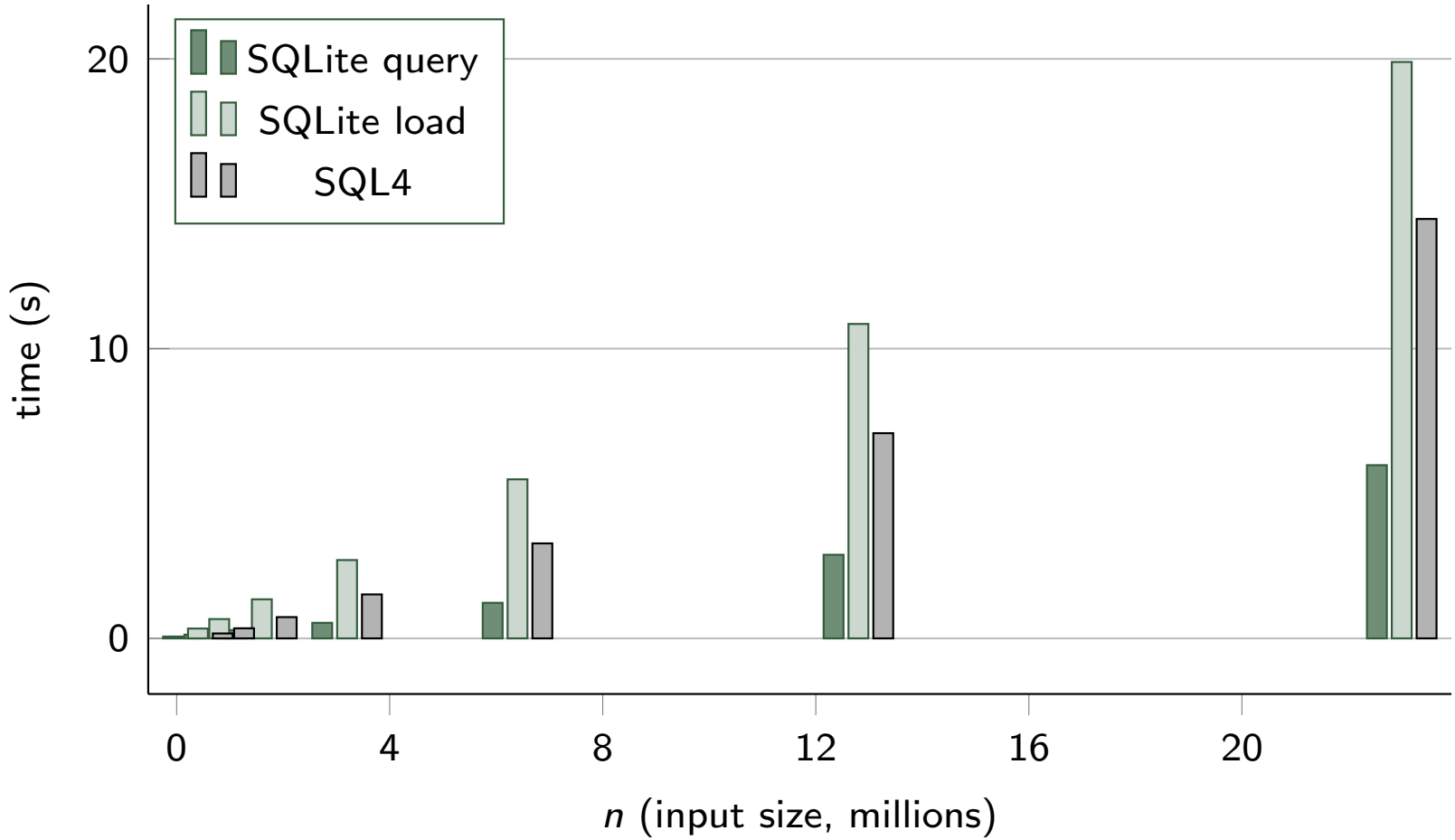
SQL queries and SQL4 functions

Rows: (*key*, *payload*) pairs with *key* uniform on $[0, 1000)$;
`eKey = NatE 999` (equivalence on keys)
`eRow = MapE fst (NatE 999)` (equivalence on whole rows)

SQL query	**SQL4 (`src/SQL4.hs`) function**
`SELECT * FROM r, s` `WHERE r.key = s.key`	`select ((fst,fst) ‘Is‘ eKey) (r ‘X‘ s)` *(dispatches to* `djoin`*)*
`SELECT key, COUNT(*)` `FROM r GROUP BY key`	`perform (Func count) (groupBy eRow r)`
`SELECT DISTINCT key FROM r`	`coalesceBy eKey (perform (Func fst) r)`

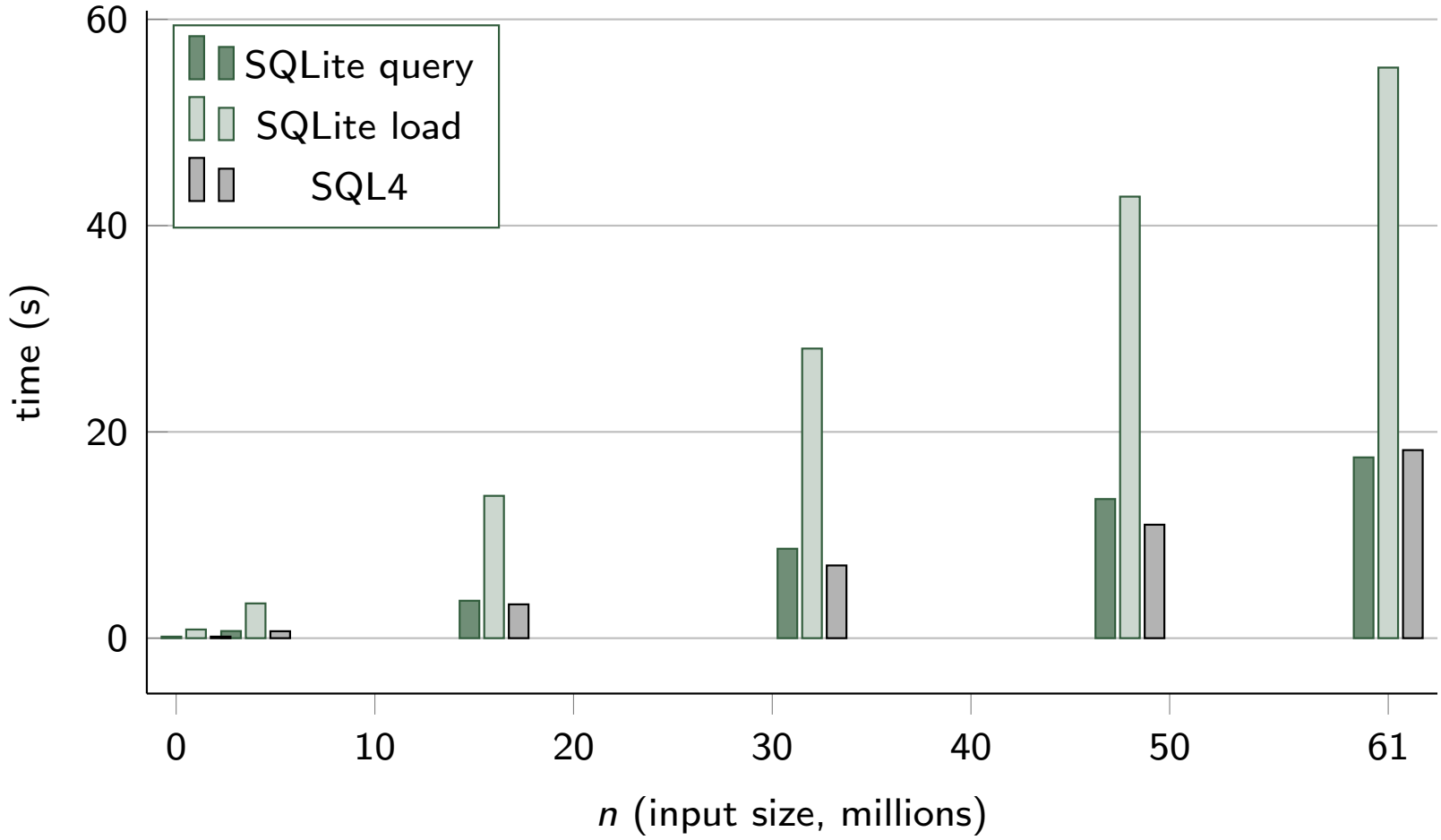


Performance: equi-join



`SELECT * FROM r, s WHERE r.key = s.key` (cardinality only: enumerating the result is intractable, e.g. 5.29×10^{11} matches at $n = 23{,}000{,}000$)

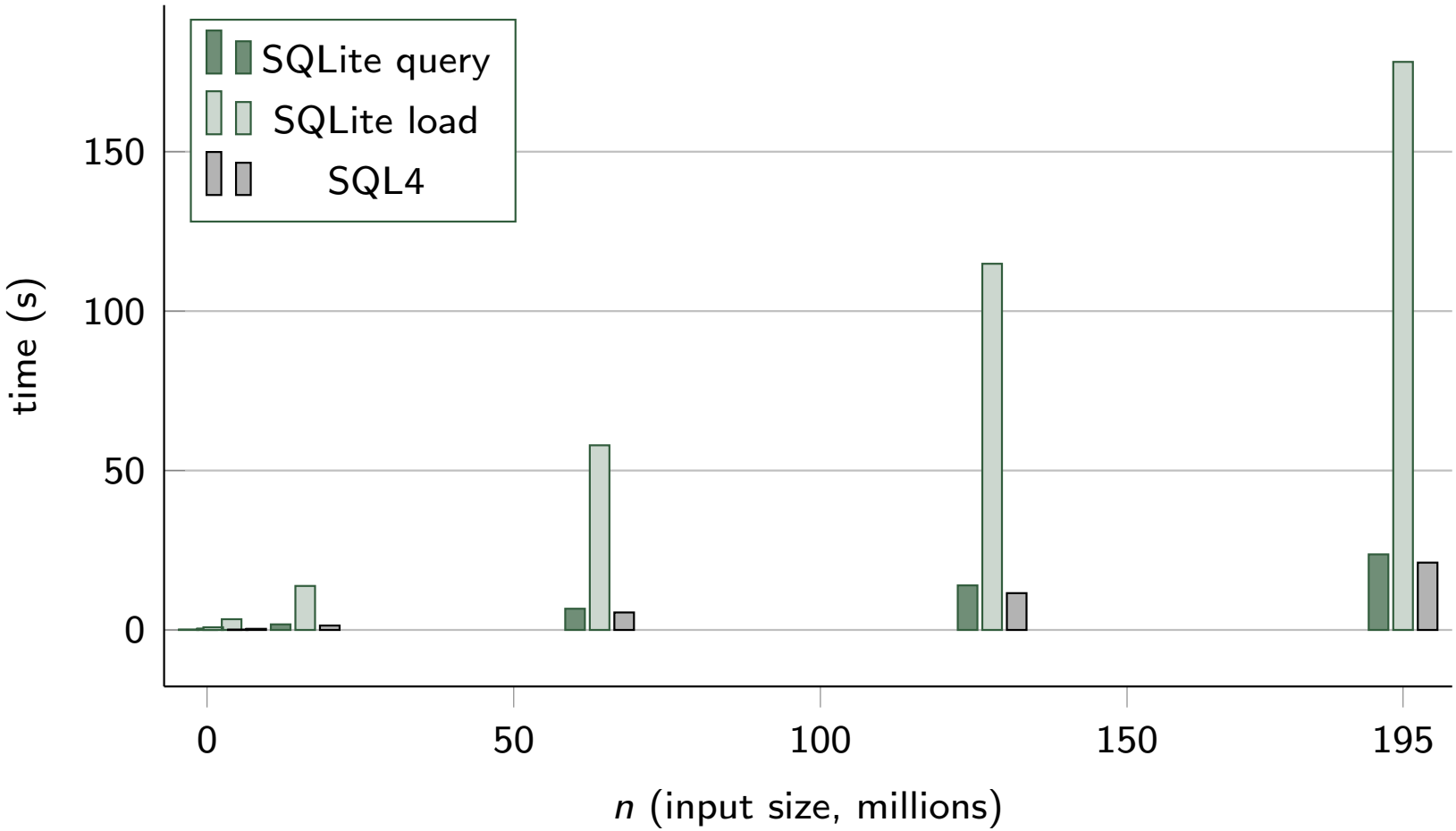
Performance: group by + count




```
SELECT key, COUNT(*) FROM r GROUP BY key
```



Performance: distinct




```
SELECT DISTINCT key FROM r
```